

GIS + Blockchain + Cybersecurity

Comprehensive Government Meeting Notes

Complete Project Flow

Drone Survey → High Resolution Images → Orthomosaic →
GIS Digitization (Properties, Roads, Drains, Water Bodies) →
Property ID Assignment → PostgreSQL + PostGIS →
Cybersecurity Protection → Blockchain (Hedera Hash) →
GIS Dashboard



1. GIS

Definition:

GIS (Geographic Information System) manages and displays location-based information.

Detailed Flow:

1. Drone captures thousands of images.
2. Images are stitched into an Orthomosaic.
3. GIS specialists digitize property polygons, roads, drains, ponds, railways and other assets.
4. Each property receives a unique Property ID.
5. Property attributes (owner, address, tax, area, mobile etc.) are stored in PostgreSQL.
6. Spatial geometry (polygon coordinates) is stored using PostGIS.
7. Clicking a polygon on the GIS map searches by Property ID and displays property details.

2. PostgreSQL + PostGIS

Why PostgreSQL?

- Reliable relational database.
- Stores owner details, tax, address, survey data.
- Supports backups and transactions.

Why PostGIS?

- Stores GIS geometries.
- Supports spatial queries.
- Stores polygons, roads, drains and coordinates.

Security:

- Authentication
- Role Based Access Control
- SSL/TLS
- Private network
- Audit logs
- Backups

3. Blockchain (Hedera)

Purpose:

Blockchain verifies integrity; it does not replace the database.

Create:

Property -> Generate SHA-256 Hash -> Send Hash to Hedera -> Transaction ID returned.

Update:

Property Updated -> New Hash -> New Hedera Transaction -> New Transaction ID.

One property can have multiple blockchain transactions:

TX001 Create

TX002 Area Update

TX003 Owner Update

TX004 Tax Update

Blockchain stores hashes and transaction history.

Application audit logs store old value, new value, officer, reason and timestamp.

4. Cybersecurity

Goal:

Protect confidentiality, integrity and availability of citizen data.

Protection Layers:

- Authentication
- Authorization (RBAC)
- HTTPS encryption
- Secure PostgreSQL access
- Firewalls
- Audit logs
- Monitoring
- Regular backups

Cybersecurity protects against unauthorized access, data theft and malicious changes.

5. Difference

GIS = Shows where the property is.

PostgreSQL = Stores property information.

PostGIS = Stores map geometry.

Blockchain = Verifies integrity and keeps tamper-evident history.

Cybersecurity = Protects the whole system.

6. Common Questions

Q: Why blockchain?

A: To verify important records have not been secretly modified.

Q: Why PostgreSQL?

A: It securely stores structured and GIS data.

Q: Can blockchain recover property data from a hash?

A: No. Hashes are one-way. The application uses Property ID and audit logs.

Q: How do you know what changed?

A: Audit logs show the details; blockchain verifies the integrity of each version.

7. 2-Minute Presentation

Our solution begins with a drone survey that captures high-resolution imagery. The images are stitched into an orthomosaic. GIS experts digitize properties, roads and drains and assign a unique Property ID. Property information is stored in PostgreSQL while spatial geometry is stored in PostGIS. We protect the application using authentication, role-based access, HTTPS, secure database access, audit logs and backups. Whenever important property information is created or updated, a SHA-256 hash is generated and recorded on the Hedera blockchain. This creates a tamper-evident history while the detailed records remain securely stored in PostgreSQL.